

A plain-language summary

of “Rule reconstruction from a single spacetime diagram yields simulation-limited prediction of the finite-horizon damage response of cellular automata”

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Which version this summarizes. This document describes the manuscript as it stands on 21 August 2026: the revision responding to the fifth referee report (registered protocol revisions 2–14; git branch `review-round7-fixes`, commit `8bf5b4a`; 16-page manuscript prepared for *Chaos*). Every number quoted below appears verbatim in that version. If the manuscript is revised again, this summary describes the older text.

1. The whole paper in one breath

We ask whether a neural network that looks at a single picture of a simple simulated universe can predict how that universe reacts to a tiny disturbance—and we show that, whenever the picture is clean, an old-fashioned and fully transparent alternative beats it: read the universe’s rule straight off the picture, then re-run the disturbance experiment yourself. The learned models only start earning their keep once the picture is corrupted by noise or missing data, and the paper maps exactly where that happens.

2. Why should anyone care?

There is a growing scientific practice of training machine-learning models to read dynamical properties—chaos, stability, sensitivity to perturbation—directly off a single observation of a system, skipping the experiments that define those properties. Published comparisons in that literature often lack strong classical baselines. This paper builds the strongest possible baseline *exactly*, in the cleanest laboratory available, works out what “as good as possible” even means when the correct answers are themselves noisy measurements, and then shows that the learned shortcut falls far short of that baseline while looking deceptively good on the usual score.

So what? Before trusting a learned shortcut for predicting how a system responds to perturbation, check whether the observation already contains the system’s mechanism. If it does, reading the mechanism out and simulating is the baseline to beat—and in this study, across every configuration tested, nothing beat it.

3. The background you need (five minutes)

Cellular automaton (CA). A row of 127 cells arranged in a ring, each cell either black or white. A fixed local *rule* decides each cell’s next colour from its current neighbourhood. The paper uses two rule families: *elementary CA*, where a cell sees itself and one neighbour on each side (256 possible rules, of which 88 are genuinely different once mirror-image and colour-swapped versions are identified—each such family is called an *orbit*), and *radius-two CA*, where a cell sees two

neighbours on each side (about four billion possible rules, from which 800 are sampled). Despite their simplicity, these systems produce everything from frozen order to full chaos to intricate interacting structures.

Spacetime diagram. Stack 127 successive rows on top of each other and you get a 127×127 black-and-white image: the complete visual history of one run. This single image is the only input every predictor in the paper is allowed to see.

Rule versus behaviour. The field distinguishes the *genotype*—the local update rule, a small lookup table—from the *phenotype*—the visible behaviour in the diagram. A key subtlety: a clean diagram contains, hidden in its pixel-to-pixel transitions, the entire rule table. Any method meant to judge *behaviour* can secretly succeed by reading the *rule* instead.

Damage spreading (the “butterfly-effect” probe). Run the same CA twice from the same starting row, but flip a single cell in one copy, and watch how the difference between the two runs evolves. Four numbers summarize the outcome: *damage survival* (does any difference remain at the end?), *damage fraction* (how much of the ring differs?), *spreading rate* (how fast does the damaged region grow?), and *cone fill* (how densely damaged is that region?). Each number is an average over 256 repetitions with random starting rows—so each “correct answer” is itself a noisy measurement, like a lab value with error bars.

The task. These four statistics are defined by *twin* runs, so no formula computes them from one image. The machine-learning proposition (“amortization”) is: pay once to train a model, then predict the four numbers from a single image, instantly, forever after.

The scores. R^2 is the standard accuracy score: 1 is perfect, 0 means no better than always guessing the average, negative means worse than that. Because the correct answers are noisy, even a perfect method cannot reach 1—an issue the paper takes very seriously. It therefore also reports ρ : the prediction error measured in units of the answer’s own measurement noise. On that scale, $\rho = \sqrt{2} \approx 1.41$ means “as accurate as re-measuring the answer with a fresh simulation” and $\rho = 1$ means “perfect knowledge of the ideal noise-free answer”.

4. A walk through the paper

Abstract, opening quotation, and Introduction (§I)

What it does. Sets up the question and announces the three results: (1) a single clean diagram identifies the exact rule for 100% of elementary and 97.5% of radius-two test rules, and re-simulating from the recovered rule is then *provably* as good as re-measuring the answer; (2) two trained direct predictors fall 3–35 noise units short of that; (3) the mechanistic route is brittle—modest noise, heavy masking, or extreme starting densities break it, and the paper maps what to use beyond.

So what? The intellectual move is taking an “obvious” trick seriously. Reading a CA rule out of its diagram is thirty-five-year-old prior art (Richards et al. 1990; Adamatzky’s 1994 monograph)—the paper claims no novelty for it. What nobody had measured is what that old fact *does* to a modern machine-learning benchmark. Answer: it quietly solves it.

The benchmark (§II): rules, targets, and what “perfect” means

What it does. Fixes the observation protocol precisely (ring of 127 cells, 127 time steps, coin-flip random starting rows, diagrams that are complete, noiseless, binary, and of known neighbourhood size), defines the four damage statistics with their edge cases, and—the section’s real content—distinguishes three different “ceilings” a predictor can be held to. Because each stored answer is one noisy measurement (estimated from $K = 20$ independent repeat measurements per rule): a predictor of the ideal noise-free answer is capped by one ceiling (the “ICC”), a predictor that is

itself a fresh noisy simulation is capped by a strictly lower one (replicate agreement), and confusing the two misstates a method by more than the margins being argued about. The ρ score compresses this into one number with fixed reference values ($\sqrt{2}$ and 1) for every target.

So what? Without this calibration, “our model scores 0.88” is uninterpretable—maybe 0.88 is nearly perfect, maybe it is far off. With it, every score in the paper can be read as a distance from what is actually achievable. This reliability machinery is reusable in any study whose ground truth comes from simulation.

Limits. The reference values $\sqrt{2}$ and 1 are exact only *on average* under an idealized noise model. The paper prices the wobble: at most 0.14 ρ -units from estimating noise with $K = 20$ repeats, up to about 0.6 from the finite rule panel—small against the 1.6–34-unit gaps the paper is about.

The three contestants (§III)

What it does. Describes the estimators. (i) *Mechanistic*: tabulate every pixel transition in the diagram, rebuild the rule table, re-run the twin experiment on the rebuilt rule. Fully transparent, no training, but it *assumes* the observation model holds. (ii) *Five cheap statistics*: five simple hand-picked image summaries (density, activity, compressibility, and two entropies) fed to gradient-boosted decision trees. (iii) *A small convolutional neural network* (CNN, 31,524 parameters) trained on many diagrams, with a first layer designed to discourage rule-reading, five training repeats with different random seeds, and—new in this version—weights chosen at the best epoch on an inner slice of training data rather than simply the last epoch (a registered protocol upgrade). The section also fixes the statistical rulebook in advance: what counts as one method being “superior”, “equivalent”, or the comparison being “inconclusive”.

So what? The contest is honest by construction: verdict rules, margins, and test panels were committed to a timestamped repository before the measurements they judge.

Result 1 (§IV A): rule reconstruction is as good as re-measuring

What it does. Shows the mechanistic route recovers the exact rule for 100% of elementary and 97.5% of radius-two held-out rules, and proves that, once the rule is exactly recovered, its predictions and a fresh independent measurement of the answer are statistically *the same thing*—so hitting the replicate ceiling is arithmetic, not luck. Scored against a $16\times$ less noisy reference answer it rises to the top ceiling, confirming its residual error was the comparison’s noise, not its own. The direct estimators do not rise when rescored the same way: their shortfall is genuine. In headline numbers: median R^2 of 0.991 (mechanistic) versus 0.880 (CNN) and 0.845 (trees) on the radius-two panel—which sounds close until converted to noise units: ρ of 0.7–1.8 versus 3–35. Two deflating findings follow. The CNN’s seemingly excellent elementary-CA spreading-rate score ($R^2 \approx 0.98$) is $\rho = 20$: twenty times the noise floor. And on damage survival, most of the direct estimators’ apparent skill is *mode assignment*: they correctly sort rules into “dies out” versus “spreads” (the CNN gets 0.94 of elementary orbits and 1.00 of radius-two rules right), but among the genuinely uncertain middle rules their R^2 collapses (CNN -1.05 on elementary CA—worse than guessing the group average). They are classifiers that had been read as regressors. The section also discloses that the radius-two test panel deliberately over-represents interesting rules, and shows by decomposition that this works *against* the paper’s own headline claim, not for it. The 2.5% reconstruction failures are examined and shown to be dynamically harmless.

So what? This is the paper’s central measurement: under clean observation, system identification is simulation-limited—as good as re-running the experiment—and the learned shortcuts are not remotely close, even where their conventional scores look flattering.

Result 2 (§IV B): the two learned methods do not even beat each other

What it does. Pre-registered paired comparisons show no stable ranking between the CNN and the five-statistic trees: the CNN is ahead on survival, the trees are superior on cone fill, the rest is inconclusive. The mechanistic estimator meets the registered superiority bar against both on most targets (differences up to +0.39 and +0.53) and loses to neither on any target in either rule space.

So what? If the expensive learned model and five cheap statistics trade places depending on the target, neither has captured the underlying physics; the method that read the rule has.

Result 3 (§IV C): what the network knows, and the memorisation test

What it does. Two diagnostics. *Stacking*: does the CNN carry information the five statistics miss? Yes, on survival (R^2 increment +0.33 over a simple base, +0.09 over the strong base)—but a deployment-realistic combination cannot harvest it (increment +0.001, not significant). *Retrieval*: a lookup that simply returns the answer of the most similar *training* rule—no learning, no simulation—matches the trained models (0.864 and 0.883 versus the CNN’s 0.880; on elementary CA the lookup inside the CNN’s own internal representation (0.883) actually beats the CNN’s own output (0.847)).

So what? Much of the learned models’ apparent skill is recognising which training rule a diagram resembles—memorisation, not generalisation to genuinely new rules. This cheap lookup test is portable to any benchmark whose space of systems is densely sampled, and is one of the paper’s most exportable tools.

Result 4 (§IV D): the rule is readable everywhere, including inside the network

What it does. Linear probes decode the exact rule identity from the CNN’s internal 64-number summary with 0.87–0.95 accuracy (versus near-zero chance), despite the anti-rule-reading first layer; individual rule-table bits transfer to never-seen rules at 0.79/0.61.

So what? The information leak the paper warns about is not hypothetical: it survives into the network trained to predict behaviour. The probes show the information is *present*; whether the network’s predictions causally *use* it is explicitly left open.

Result 5 (§IV E–F): where reading fails, and what wins beyond

What it does. Deliberately breaks each observation assumption and maps the consequences on a grid (the “identifiability frontier”). Exact reconstruction is brittle exactly where you would expect: a deterministic rule table has no concept of noise, so a single corrupted pixel transition can flip a table entry—at 5% pixel noise the mechanistic score falls below zero. It tolerates about 25% missing pixels and fails at extreme starting densities. Then comes the repair race between eight registered estimators plus two fairness controls: a *probabilistic* rule reader (the “pseudo-posterior decoder”—keep a per-entry probability instead of a hard bit, simulate several plausible tables, average) extends the working range to 50% masking (R^2 0.85 at 40% masking versus the exact inverter’s 0.24) and rescues extreme densities (0.77 versus 0.08), but only helps to about 3% noise, and its confidence statements become badly overconfident under noise (nominal 68% intervals actually cover 58% at 2% noise and 13% at 20%). A learned rule-reading network mostly fails—a memorable finding along the way: a table that is 73% correct produces a *badly wrong* dynamical system, because a rule table is not the kind of object where “mostly right” helps. The fairness controls matter most: retraining the same CNN *on corrupted diagrams* makes it the best method in the 3–15% noise band (R^2 0.51–0.62)—so an earlier “dead zone” where nothing worked was an artifact of testing clean-trained models—while an unconstrained network 354× larger changes no verdict, and a control that puts the interesting rules back into training still leaves the CNN at 0.778 versus the mechanistic 0.992. At 20% noise nothing tested is reliable. All of this is condensed

into a practitioner guidance table: which estimator to use in which observation regime, with its price.

So what? This is the transferable product. The matched-regime verdict (“just read the rule”) is specific to clean CA diagrams; the *frontier*—exact methods first, probabilistic reading under moderate corruption, corruption-trained networks beyond, nothing past heavy corruption—is the pattern the authors expect to generalize, and each axis has a named analogue in real system identification (sensor noise, missing sensors, insufficient excitation).

Secondary analyses (§IV, final subsections): cost, and honest by-products

What it does. *Cost:* the mechanistic estimator pays per question (283 ms) versus the CNN’s 1 ms, but training means the CNN only breaks even after ≈ 4700 questions (plausibly ≈ 50 against an optimized implementation)—and even a bargain-budget mechanistic run (17 ms) still beats both learned models on accuracy. A zero-cost textbook approximation (the “annealed” mean-field theory) fails completely ($R^2 = -2.15$), for an identified reason: the targets measure precisely the correlations that approximation throws away. *By-products:* a map of the radius-two rule space in damage coordinates (7.8% of rules meet a registered “interesting damage” criterion— explicitly *not* promoted to a validated census of complex rules), and a sliding-window “behaviour map” for mixed-rule CA whose advantage is statistically inconclusive. Both are reported at exactly the strength their evaluation supports.

So what? Speed is the only case left for the learned models in the matched regime, and it is weaker than it looks. The by-products show the flip side of the paper’s discipline: results that did not pan out are reported as such rather than dressed up.

Discussion (§V)

What it does. States the claim at its correct scope: a single-observation behaviour-prediction benchmark is solved by system identification *whenever* the observation pins down the generator within a known model class, the inversion is tractable, and the defining experiment can be re-run. Benchmark designers who intend to test representation learning must break that chain (noise, partial observation, unknown model class)—or accept that they are benchmarking identification. The authors are careful about reach: they name no published benchmark this invalidates, and they name their own most consequential omission—the fully correct probabilistic decoder for noisy diagrams was not built, so the noise-regime verdicts bound the estimators tested, not the whole mechanistic family. The identified next experiment: stochastic CA (Domany–Kinzel), where the rule itself is probabilistic, identification becomes genuine estimation, and the paper’s near-identity argument should finally break down.

So what? The discussion converts one clean case study into a design rule for a whole literature, while drawing the boundary of its own evidence unusually sharply.

Appendices A–E

A: Reproducibility and registration. The full ledger of which analyses were registered before measurement, which were added afterwards, and which are exploratory; plus a correctness audit of the central identity and one disclosed deviation from the original registration. *B: Methods supplement.* Exact data, splits, network architecture, evaluation protocol, and the record of the checkpoint-selection switch (the registered rule fired: radius-two cone fill rose $0.337 \rightarrow 0.406$ and seed-to-seed spread collapsed $0.109 \rightarrow 0.036$). *C: Decoder algorithms.* Pseudocode for every rule-reading variant, with a worked example showing why the probabilistic decoder becomes overconfident. *D: The damage-signature landscape.* The rule-space map with its central caveat spelled out: the striking-looking region is substantially a coordinate effect, a registered threshold

region rather than a discovered cluster; an intended independent detector performed at chance and validates nothing. *E: Prior work under its assumptions.* A table of thirty-five years of CA identification literature, so the boundary of the novelty claim is auditable.

5. The limits, stated plainly

- **Everything is protocol-scoped.** All statistics are defined for this exact setup: ring of 127 cells, 127 steps, coin-flip starting rows. This is deliberate (the asymptotic versions are formally undecidable), but it means no claim is about “the rule” in the abstract.
- **The neighbourhood size is assumed known** in the main analysis; inferring it costs about 0.12 in median R^2 .
- **The noise-regime verdicts bound the tested estimators only.** The correctly specified probabilistic decoder for noisy diagrams was not built; the authors flag this as their most consequential omission.
- **Neural claims are scoped to this architecture and budget.** No architecture search was run; the paper explicitly does not conclude that deep networks cannot close the gap.
- **The radius-two test panel is enriched** with interesting rules selected using the same data that supplies the evaluation labels—disclosed, and shown by two controls to be conservative for the headline claim, but a design wart nonetheless.
- **Deterministic, binary, one-dimensional CA only.** The stochastic case is argued to be the thesis’s breaking point—and has not been run.
- **The frontier rests on a fixed 80-rule panel;** only the noise axis was replicated on a second panel (the qualitative structure held, and the crossover moved—it is a band, not a line); corruptions were tested one axis at a time.
- **Timings are one machine, one implementation.**

6. Why the results are valuable anyway

A calibrated negative result. Most of the paper’s value is in what it stops people from doing: publishing learned single-observation predictors of perturbation response without building the identification baseline. The gap is not asserted but measured, in units that make it unmistakable.

Reusable measurement machinery. The three-ceiling reliability analysis and the ρ score apply to any benchmark whose ground truth comes from noisy simulation—which is most of simulation-based machine learning.

Two cheap, portable diagnostics. The mode-assignment check (is the regressor secretly a classifier?) and the nearest-training-example lookup (is the model secretly memorising?) can be run on any benchmark in an afternoon.

A practical map. The frontier and its guidance table tell a practitioner which estimator to trust for a given observation quality, with measured evidence and explicit refusal to recommend anything where nothing worked.

A methodological standard. Decision rules were frozen before measurements, every quoted number is re-derived from released artifacts by an audit script, negative and inconclusive results are reported at face value, and each claim is labelled registered, post-hoc, or exploratory. The paper practises the benchmark hygiene it preaches.

This summary accompanies the repository copy of the manuscript. It simplifies wording but adds no claims; where it paraphrases, the manuscript's own hedges apply. Source of record: `manuscript/main.tex` at commit `8bf5b4a`.